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Abstract: This contribution proposes the texts to advance the Recommendation ITU-T Q.CEIR for the technical requirement, interfaces and generic functions of CEIR.

~~Recommendation~~ Recommendation ITU-T Q.CEIR

Technical requirement, interfaces and generic functions of CEIR

Summary

With the expansion of mobile devices communication adoption, the world is facing increasing cases of theft mobile devices theft, replication of IMEI, and the use of counterfeit mobile devices. To prevent usage of these irregular devices, one of the possible actions recommended at ITU-T Q.5050, ITU-T Q.5051 and ITU-T Q.5052 is the Mobile Network Operator (MNO) block the irregular mobile device unique identifier at the Equipment Identity Register (EIR) and implement of a series of device control lists (such as Permitted/White, Grey and Restricted/Black) to manage this process.

The ITU-T Q.5050, ITU-T Q.5051 and ITU-T Q.5052 also indicates the need for an irregular device to be blocked at the same time in different MNOs (within the country and also around the world) and for that is required the use of a Central Equipment Identity Register (CEIR) that consolidate and distribute the data contained on these control list to all the connected Mobile Network Operator (MNO).

To assist the countries aiming to implement a Central Equipment Identity Register (CEIR), this recommendation provides detailed technical description of the CEIR system, its requirements, interfaces and basic functions that should be provided by the system and also some optional features – which may be used considering the client specific requirements. It also describes the different stakeholders of the CEIR system and their indented/expected roles and responsibilities.

History

Edition	Recommendation	Approval	Study Group	Unique ID*
1.0	ITU-T Q.CEIR	---	11	--

Keywords

CEIR, Combat counterfeit ICT and stolen devices, unique identifiers, replicated devices

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As of the date of approval of this Recommendation, ITU had not received notice of intellectual property, protected by patents, which may be required to implement this Recommendation. However, implementers are cautioned that this may not represent the latest information and are therefore strongly urged to consult the TSB patent database at <http://www.itu.int/ITU-T/ipr/>.

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~~Recommendation~~Recommendation ITU-T Q.CEIR

"Technical requirement, interfaces and generic functions of CEIR"

1 Scope

This Recommendation proposes the requirement of a CEIR (Central Equipment Identity Register) and the generic features and interfaces which should be implemented.

2 References

This ITU-T Recommendation and other references contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Recommendation are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below. A list of the currently valid ITU-T Recommendations is regularly published. The reference to a document within this Recommendation does not give it, as a stand-alone document, the status of a Recommendation.

ITU-T Recommendation Q.5050, "Framework for solutions to combat counterfeit ICT devices"

ITU-T Recommendation Q.5051, "Framework for Combating the use of Stolen Mobile Devices"

ITU-T Recommendation Q.5052, "Addressing mobile devices with a duplicate unique identifier"

ITU-T Recommendation Q.5053, "Mobile device access list audit interface"

3 Definitions

3.1 Terms defined elsewhere

This Recommendation uses the following terms defined elsewhere:

blacklist override (BLO) [ITU-T Q.5053]: The list of subscribers that have blacklisted mobile devices (blocked international mobile equipment identities (IMEIs)), but who are still allowed to access mobile networks and services.

counterfeit ICT device [ITU-T Q.5050]: An information and communication technology (ICT) device that explicitly infringes the trademark, copies hardware or software designs, or infringes brand or packaging rights of an original or authentic product and, in general, infringes applicable national and/or international technical standards, regulatory requirements or conformity processes, manufacturing licensing agreements, or other applicable legal requirements.

equipment identity register (EIR) [ITU-T Q Suppl. 73]: An EIR is a network element in the core of mobile networks that is used to terminate an access attempt or ongoing call when performing IMEI Check procedure depending on the status of the IMEI in one of its registers; blocked-list, permitted-list, or tracked-list.

mobile device identifier database (MDID)[ITU-T Q.5052]: A database containing aggregated information about mobile device unique identifiers.

smartphone [ITU-T X.Sup.19]: A mobile phone with powerful computing capability, heterogeneous connectivity and advanced operating system providing a platform for third-party applications.

tampered ICT device [ITU-T Q.5050]: An information and communication technology (ICT) device that had components, software, unique identifier, items protected by intellectual-protected rights or trademarks tentatively or effectively altered without the explicit consent of the manufacturer or its legal representative.

unique identifier [ITU-T Q.5050]: An identifier associated with a single device that aims to uniquely identify it.

3.2 Terms defined in this Recommendation

This Recommendation defines the following terms:

None.

4 Abbreviations and acronyms

This Recommendation uses the following abbreviations and acronyms:

API	Application Programming Interface
ATM	Automated Teller Machine
BL	Blocked-list
BLO	Blocked-list override
EIR	Equipment Identity Register
GSMA	Global System for Mobile Communications (GSM) Association
ICT	Information and Communications Technology
<u>IDS</u>	<u>Intrusion Detection Systems</u>
IMEI	International Mobile Equipment Identity
IMSI	International Mobile Subscriber Identity
<u>IPS</u>	<u>Intrusion Protection Systems</u>
<u>LEA</u>	<u>Law Enforcement Agency</u>
MDID	Mobile Device Identifier Database
MNO	Mobile Network Operator
MSISDN	Mobile Station Integrated Services Digital Network
ODM	Original Design Manufacturer
OEM	Original Equipment Manufacturers
PIN	Personal Identification Number
SMS	Short Message Service
TAC	Type Allocation Code
<u>VPN</u>	<u>Virtual Private Network</u>
<u>WAF</u>	<u>Web Application Firewall</u>

5 Conventions

This Recommendation applies the following verbal forms for the expression of provisions:

- a) The keyword "is required to" indicate a requirement which must be strictly followed and from which no deviation is permitted, if conformance to this Recommendation is to be claimed.

- b) The keyword "should" indicate a requirement which is recommended but which is not absolutely required. Thus, this requirement need not be present to claim conformance.
- c) The keyword "may" indicate an optional requirement which is permissible, without implying any sense of being recommended. This term is not intended to imply that the vendor's implementation must provide the option, and the feature can be optionally enabled by the network operator/service provider. Rather, it means the vendor may optionally provide the feature and still claim conformance with this Recommendation.

6 General aspects

With the expansion of mobile devices communication adoption, the world ~~is~~ facing increasing cases of mobile devices theft, replication of IMEI, and the use of counterfeit mobile ~~devices~~. To prevent usage of these counterfeit mobile devices, one of the possible actions recommended at ITU-T Q.5050, ITU-T Q.5051 and ITU-T Q.5052 is the Mobile Network Operator (MNO) block the irregular mobile device unique identifier at the Equipment Identity Register (EIR) and implement of a series of device control lists (such as Permitted/White, Grey and Black/Blocked) to manage this process.

The ITU-T Q.5050, ITU-T Q.5051 and ITU-T Q.5052 also indicates the need for an irregular device to be blocked at the same time in different MNOs (within the country and also around the world) and for that the use of a Central Equipment Identity Register (CEIR) is required to consolidate and distribute the data contained on these control list to all the connected Mobile Network Operator (MNO).

To assist the countries aiming to implement a Central Equipment Identity Register (CEIR), this recommendation provides detailed technical description of the CEIR system, its requirements, interfaces and basic functions that should be provided by the system and also some optional features – which may be used considering the client specific requirements. It ~~is~~ also describes~~d~~ the different stakeholders of the CEIR system and their indented/expected roles and responsibilities.

6 Need for a Central Equipment Identity Register (CEIR) System

The number of mobile communication devices has shown a drastic increase over the last two decades throughout the world. With the expansion of wireless and wireline communication network infrastructure and the number and functionality of smartphones and tablets, access to individual and corporate data services has also been significantly enhanced. According to ~~the~~ International Telecommunication Union (ITU), the noticeable growth in the capacity and number of broadband wireless communication devices has introduced a serious security challenge with regard to theft of these devices as well as unauthorised access to personal data and other data security issues.

With the growing importance of mobile devices and ~~a~~ variety of new applications, in addition to voice and data communications, consumers are using them for storing a number of personal details like phone contacts, bank account details, passwords, ATM PIN, date of birth etc. Thus, mobile device has become a valuable item particularly in terms of the personal data/information stored in it.

With mobile success stor~~ies~~, issues like increasing theft cases of mobile devices, replication of mobile device unique identifier~~s~~, illegal & non-genuine mobile devices have also cropped up. Mobile devices with new technologies such as 3G/4G/5G Smart phones with advanced features and applications are expensive in the market and reselling of stolen devices becomes lucrative for thieves. Stolen mobile devices are re-programmed to change their IMEI numbers so that it cannot be traced.

To block replicated IMEIs and stolen/lost mobile devices pan country, there's a requirement for a central database of all IMEIs used in a country, categorize into Permitted, Tracked and Blocked list and eventually the Mobile Network Operator (MNO) Equipment Identity Register (EIR) can block

the Blocked listed devices in its own network. The mobile device blocked in one MNO can still be used in other MNOs. However, to prevent usage of stolen/lost devices in all regions (telecom circles) and different MNOs, a Central Equipment Identity Register (CEIR) needs to be established, which provides these consolidated blocked list data to all EIRs in the nation to ensure countrywide blocking of the stolen/lost mobile devices.

Use of CEIR/ national reference database is also recommended by ITU through following recommendations -

- a) ITU-T Q.5050, "Framework for solutions to combat counterfeit ICT devices" recommended to deploy a centralized reference database of authorized equipment in a specific market, based on unique identifiers, to effectively differentiate between genuine and counterfeit ICT devices.
- b) ITU-T Q.5051, "Framework for Combating the use of Stolen Mobile Devices". It is recommended to use a centralized reference database to store the information of lost and stolen devices. Therefore, all carriers should make use of this database to prevent stolen devices from accessing any mobile network. This database should contain, at least, the unique identifier of the stolen device, the date of the event and the entity that included the information on the database.
- c) ITU-T Q.5052, "Addressing mobile devices with a duplicate unique identifier" also recommends the same central database with different name – it says that a robust Mobile Device Identifier Database (MDID) sharing information with national MDIDs among countries (bilateral, regional or global) will aid in detecting counterfeit mobile devices distributed or sold in such a manner. The objective is to detect instances of IMEIs that are duplicate across countries. Duplicate detection of IMEIs within a country should leverage other methodologies described in this Recommendation. An MDID will be established in such a manner that after connection it is seamlessly interoperable with other national MDIDs of individual countries in a region or sub-region by sharing a list of IMEIs.

In addition to discouraging theft of the mobile device, the CEIR system will help the national security authorities by tracking the whereabouts of particular stolen/lost mobile devices throughout the coverage area/ country.

7 Description of the System

CEIR is a nationwide reference database system used to maintain database containing the IMEI (International Mobile Equipment Identity) and IMSI (International Mobile Subscriber Identity) of all the mobile devices used in the nation for the purpose of detecting replicated IMEIs and maintaining blocked-listed (BL) IMEIs centrally. Optionally it can store MSISDN (Mobile Station Integrated Services Digital Network) of all the records if there's a national requirement to send SMS notification to the subscribers. It will provide a facility to register stolen/lost mobile devices and generate Blocked list. All MNOs update their local EIR blocked list from the CEIR blocked list, preventing the use of BL mobile device across all MNOs within the nation. The stolen/lost mobile device is reported by MNO if its usage is attempted and the corresponding traceability details may be presented to the local/regional/state police for recovery. Blocked-list override (BLO) can also be implemented by CEIR to MNO EIR to provide mobile services to the subscribers that possess replicated and blocked-listed mobile devices, but still allowed to access mobile networks and services. CEIR can maintain the Device Registry of all authenticated mobile devices available in the country – which will include locally manufactured and imported to the country. The system can also provide other features like mobile device verification, authentication, device pairing etc. Additionally, it can also provide solution to cross-check and restrict import of counterfeited devices to the nation.

8. CEIR Architecture

CEIR is a central system that connects all the required stakeholders. Each stakeholder plays an important role to accomplish the goal of the CEIR system. Similarly different modules have significant roles in the system. All stakeholders and different modules are as shown below -

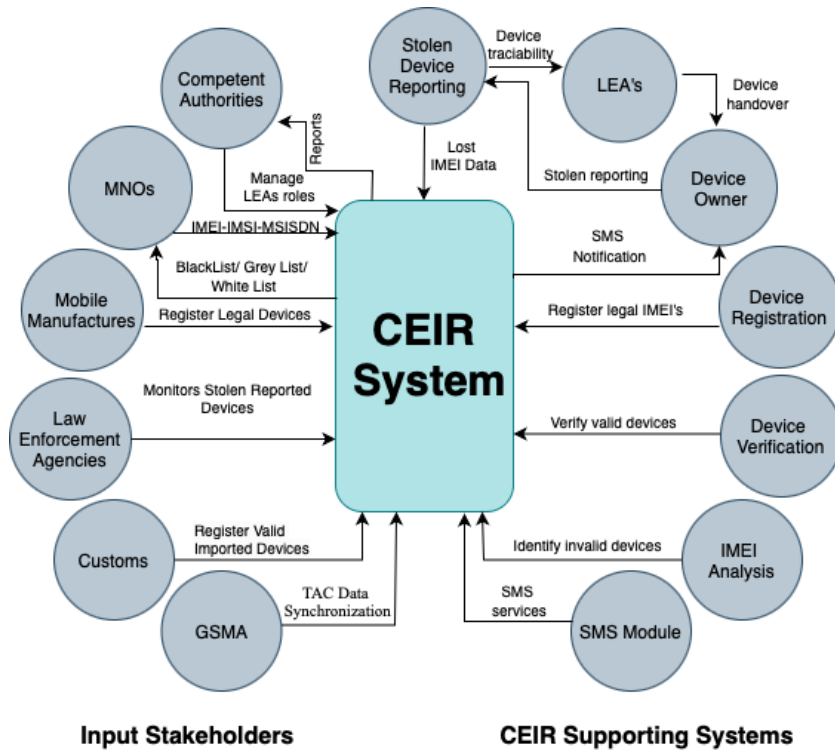


Figure : 1 CEIR architecture with stakeholders and modules

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Following is the network architecture of the CEIR system –

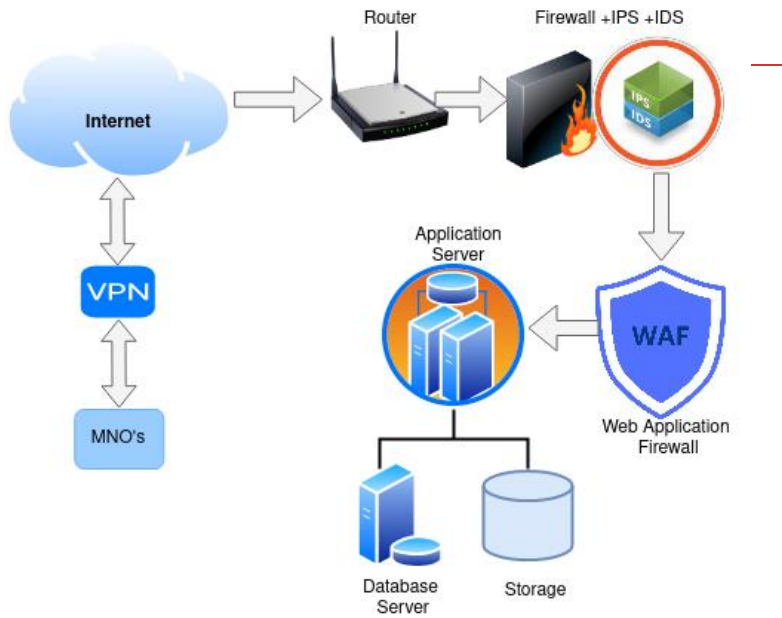


Figure 2 : CEIR network architecture

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As defined in Rec. ITU-T Q.5052, CEIR could be implemented in three levels such as National level, Regional level and Global level based on the requirements and feasibility. Multiple National level CEIRs will be connected to Regional CEIR. Similarly multiple regional level CEIRs will be connected with Global CEIR

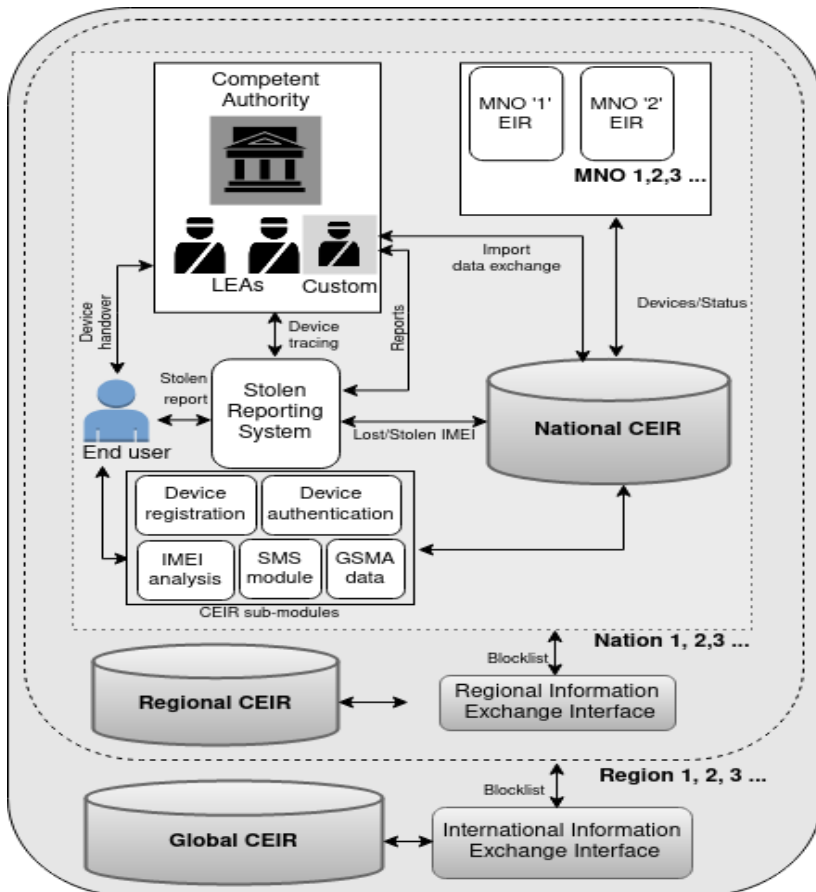


Figure 3 : CEIR multi-levelled architecture

8.1 9. ExternalSystem Interfaces

The deployment of the CEIR solution shall be a two-tiered structure where the CEIR is deployed at Central Level and the EIR are deployed in each MNO. Depending upon the interfaces and the information exchanged among the two there could be three deployment methodologies as described in ITU-T Q.Sup.CEIR-EIR-int:

- API-Based (synchronous) query interface – where the exchange of information among the CEIR and EIRs is in near real time.
- API-based (asynchronous, near real time) interface – where data exchange will be in near real time.

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- iii. File-based offline interface – where the exchange of information among CEIR and EIRs are at predefined intervals and there is lag in enforcement of the- blocking/ tracing of the IMEI.

~~As defined in Rec. ITU-T Q.5052, CEIR could be implemented in National level, Regional level and Global levels also based on requirement and feasibility.~~

10. System features

8.2 10.1 Desired System Features/ functions

Following are the desired features/functions of the CEIR system -

- 1) Receive data from all MNOs on a daily basis preferable online and upload in CEIR database. For effective functioning, real time synchronous query-based interface may be used – where real-time replicated device detection and service barring is possible. In case of any limitation, MNO data transfer could be near real time or periodic. Based on the data transfer mechanism chosen, replicated mobile device detection, blocking and service barring will be performed.
- 2) Daily receipt and updating of latest TAC data from GSMA to update CEIR for newly allocated TAC. Otherwise, the newly manufactured device can be treated as an illegal device by CEIR. As on date, GSMA updates their TAC database once daily – so daily data synchronisation is sufficient till any frequency changes done by GSMA.
- 3) Feature of online/offline device analysis, identification and replicated mobile device detection. Identification of genuine and counterfeited devices. This will be based on the data transfer mode selected.
- 4) Online/offline BL/BLO list generation for MNOs and interface to update all MNO's EIR with latest BL/BLO. Based on the available BL/BLO, MNO EIR will update their list and accordingly services of the mobile devices will be provided or barred.
- 5) Maintain Mobile Device Registry for all valid and active devices including local manufacturers, imported and international in-roamers. When complete device registry will be prepared, the list will be used to verify whether the mobile device is valid or not.
- 6) Ability to handle the- circulation and use of illegal imported devices and allow the legal and authentic imported devices.
- 7) Provision of amnesty to older existing non-compliant or illegal mobile devices, configurable as per national requirement.
- 8) Gradual reduction in the replicated/counterfeited Mobile Equipment Identity in the country's network.
- 9) Tracking the block listed mobile usage. Tracing and detection/retrieval of the lost/stolen mobile devices.
- 10) Keep a check on the entry/production of the counterfeited mobile devices.

8.3 10.2 Optional system Features/functions

Following are the desired features/functions of the CEIR system -

- 1) Utility for mobile device users and purchasers to check device validity even before purchase. This feature may be provided for the use of the mobile subscribers through SMS, web portal or through mobile App. Using this feature the subscriber will be in a position to detect whether a mobile device is genuine or not, according the subscriber will take decision whether to buy the device or not.
- 2) Portal to report Lost/ Stolen mobile devices to block-list the device in the national mobile network. It will facilitate subscribers to report their stolen/lost mobile devices.

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- 2)3) Portal to govt. agencies to update the status of stolen reported mobile devices.
- 3)4) ModuleSystem to discourage import of replicated mobile devices to the nation by verifying all the IMEIs imported to the nation.
- 5) SMS notification to the public users for several important information and tracking status of stolen reported mobile devices.
- 6) Mass mobile identity blocking/unblocking interface for the government agencies /regulators in the case when the complete consignment is stolen/ bulk IMEIs are identified for illegal activities.
- 7) Portal to the Law Enforcement Agencies to generate intelligent analysis reports based on different parameters such as location, duration, device type, make and model etc.
- 8) Utility for mobile device users and purchasers to check whether the mobile device is manufactured locally or imported, quality/security rating based on the device registry and survey results.
- 9) Interactive interface of feedbacks and grievances on stolen reporting for mobile users.
- 10) Interfacing to synchronize the older stolen devices complaints in govt. agencies e.g. police to integrate with the CEIR system.
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- 4)12)

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8.4 11 Stakeholders

- 1) The mobile device original equipment manufacturer (OEM)/ original design manufacturer (ODM)/ Importers/ Brand owners.
- 2) The Mobile Network Operators (MNOs).
- 3) The Law Enforcement Agencies and other Government agencies.
- 4) The citizen/subscribers
- 5) The Regulatory organisation

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12 Non-functional Requirements

12.1 Performance Requirements

All other performance related to storage, memory, and processing should follow industry recommended practices to ensure resource requirements are minimized.

12.2 Reliability Requirements

In accordance with industry recommended practices, the system should undergo feature testing, load testing, and regression testing prior to release and/or deployment.

12.3 Availability Requirements

Reasonable efforts should be made to ensure the software system is available with an preferable uptime of 99.99%. The uptime is calculated as the percentage of time during the year in which the software system was available to the public.

12.4 Security Requirements

The system defined in this document must follow industry recommended practices for secure software development. At a minimum, the software development must practice the principle of least privilege for defining access-level requirements of the software system and its associated services. The production-release version of the software system must pass an automated dynamic application security testing/audit tool.

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